RETAINX – CUSTOMER RETENTION & CHURN ANALYTICS

END-TO-END ANALYTICS PROJECT
PYTHON | SQL | POWER BI

UJJWAL VERMA | DATA ANALYST | SQL • PYTHON • POWER BI • EXCEL

BUSINESS CONTEXT & PROBLEM

Why This Problem Matters

- Customer churn is one of the biggest revenue risks for Telecom, SaaS, OTT & FinTech companies
- Acquiring new customers costs 5–7x more than retaining existing ones
- Businesses struggle to:
    1. Identify why customers churn
    2. Detect who is at risk early
    3. Quantify revenue impact of churn

Goal: Enable leadership to reduce churn and protect revenue using data-driven insights

PROJECT OBJECTIVES

What RetainX Solves

- Analyze customer behavior and churn patterns
- Segment customers by:
    1. Usage
    2. Tenure
    3. Revenue
- Identify high-value customers at risk
- Quantify revenue at risk
- Deliver executive-ready dashboards

This project simulates how analytics teams work in real companies.

DATA & TOOLS OVERVIEW

Data Source

1. Public Telecom Customer Dataset (Kaggle)

2. Customer-level behavioral & churn data

Tech Stack

- Database: PostgreSQL
- Data Processing: SQL
- EDA & Validation: Python (Pandas, NumPy, MatPlotlib, Seaborn)
- Reporting: Excel
- Visualization: Power BI
- Version Control: Git & GitHub

END-TO-END WORKFLOW

How the Project Was Executed

1. Business Understanding & KPI Definition
2. Data Ingestion (Raw Layer)
3. Feature Engineering (SQL)
4. Analytical Table Creation (Gold Layer)
5. Python EDA & Validation
6. Power BI Dashboarding
7. Business Insights & Recommendations

DATA ARCHITECTURE

LAYERED DATA MODEL

- Raw Layer :

  Original data ingested into PostgreSQL

- Transformation Layer :
  - Revenue segmentation
  - Usage score
  - Usage category
  - Tenure calculation

- Analytical (Gold) Layer :
  - Customer segmentation
  - Business-ready fact table
  - Optimized for Power BI

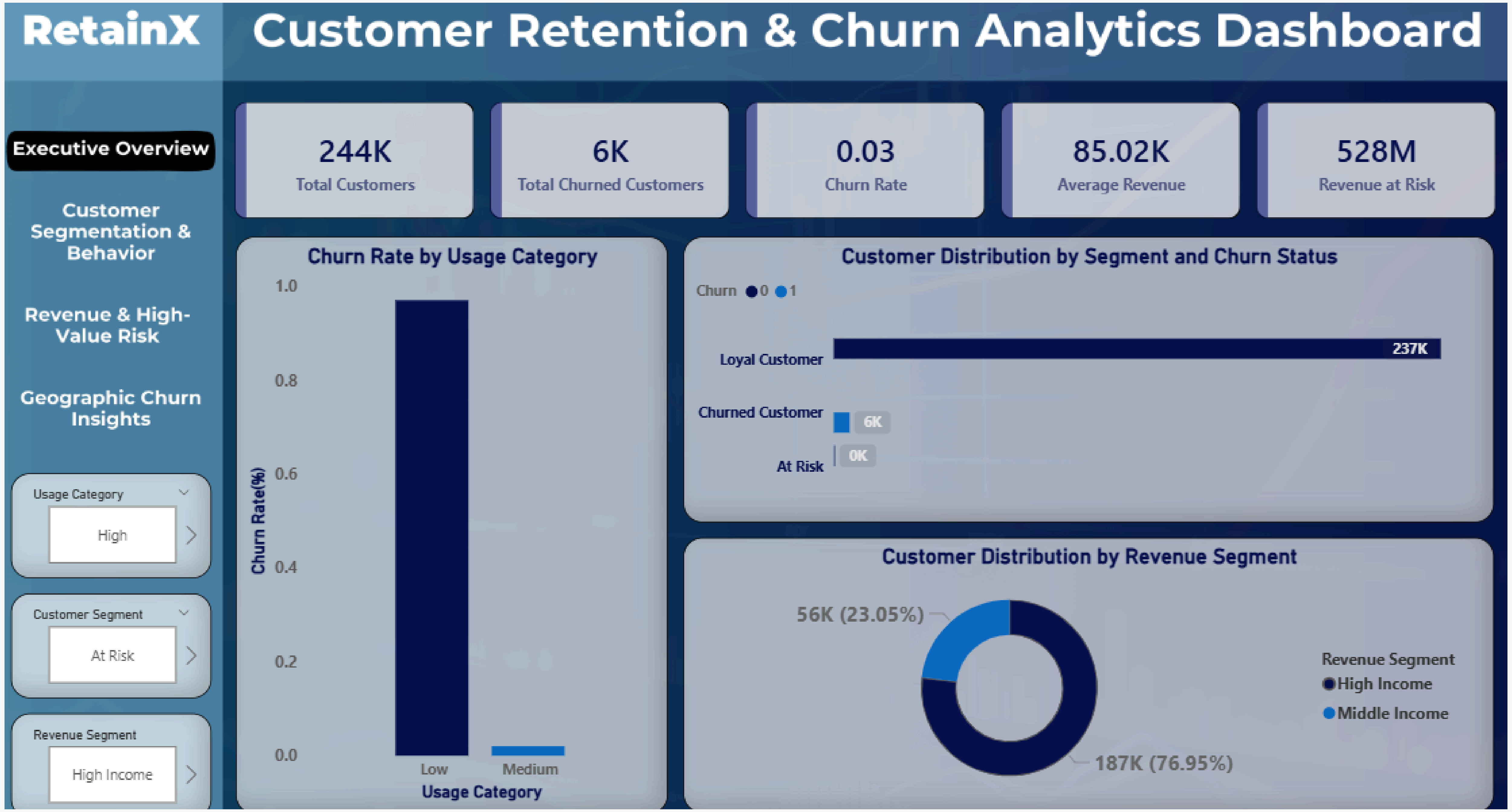
KEY KPIS DEFINED

- Business Metrics Tracked :
  1. Total Customers
  2. Total Churned Customers
  3. Churn Rate (%)
  4 .Average Revenue
  5. Revenue at Risk
  6.High-Value Customers at Risk

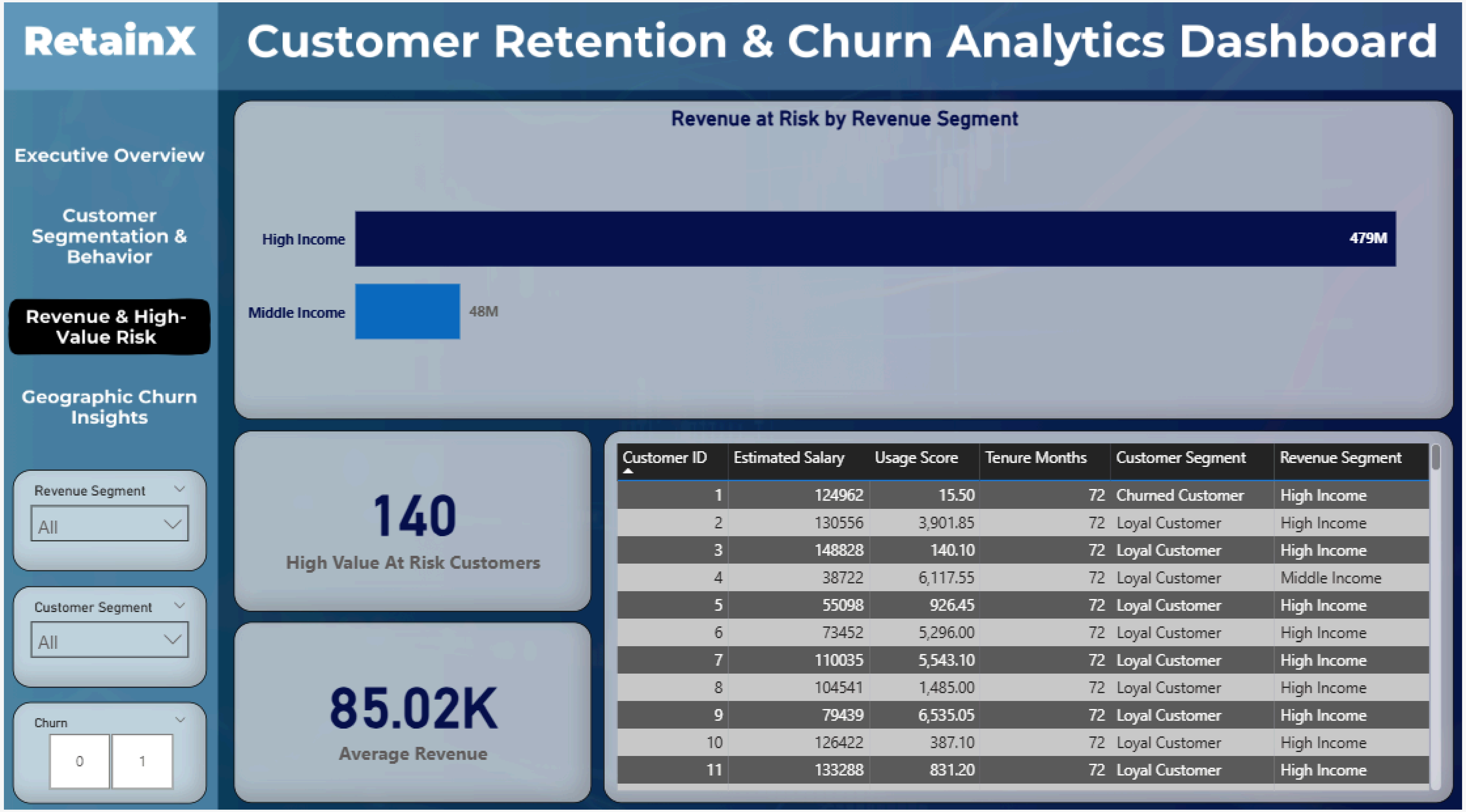
- Churn Analysis By :
  1. Usage Category
  2. Customer Segment
  3. Tenure
  4. Geography

POWER BI DASHBOARD

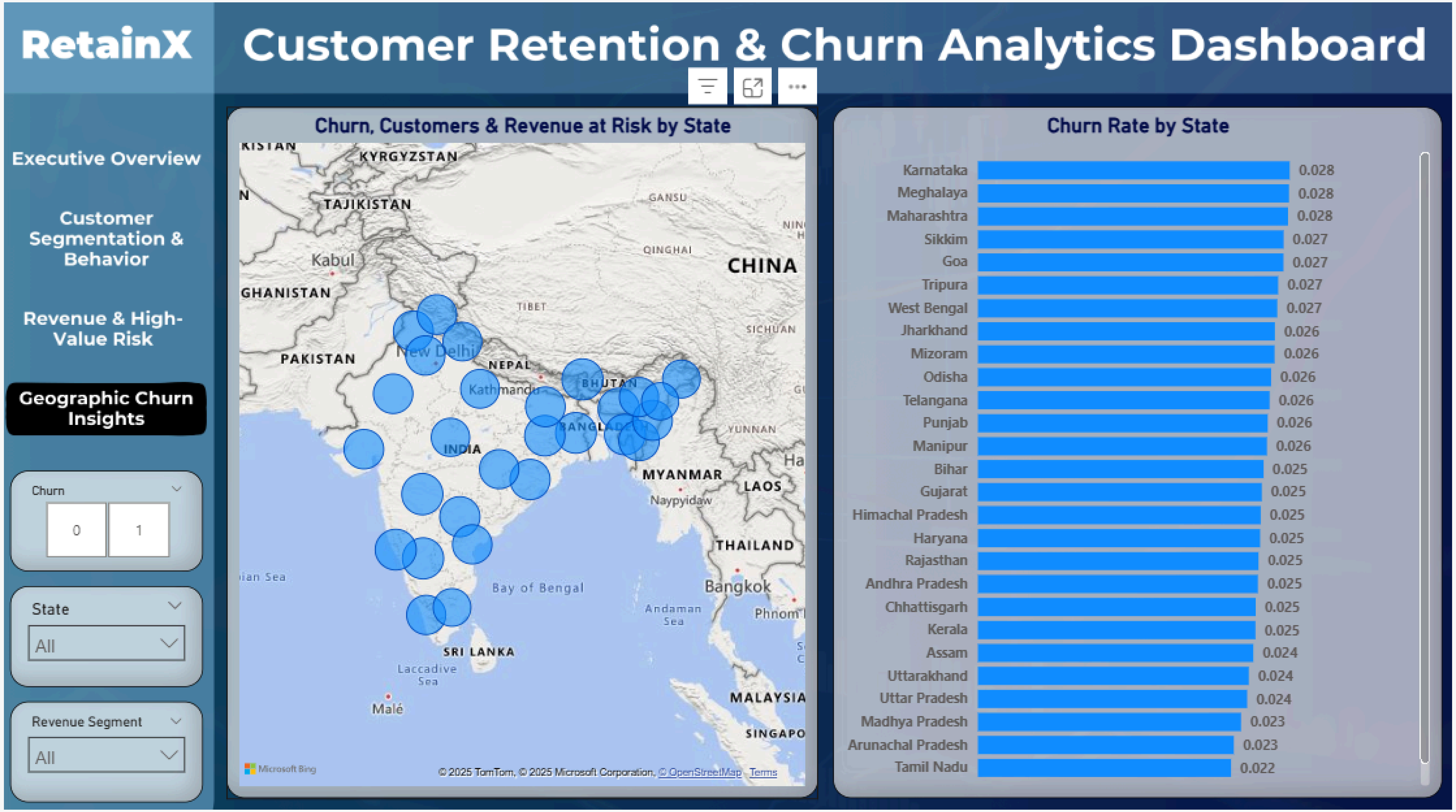
- Overall churn health of the business
- Revenue exposure at a glance
- Customer distribution by risk level



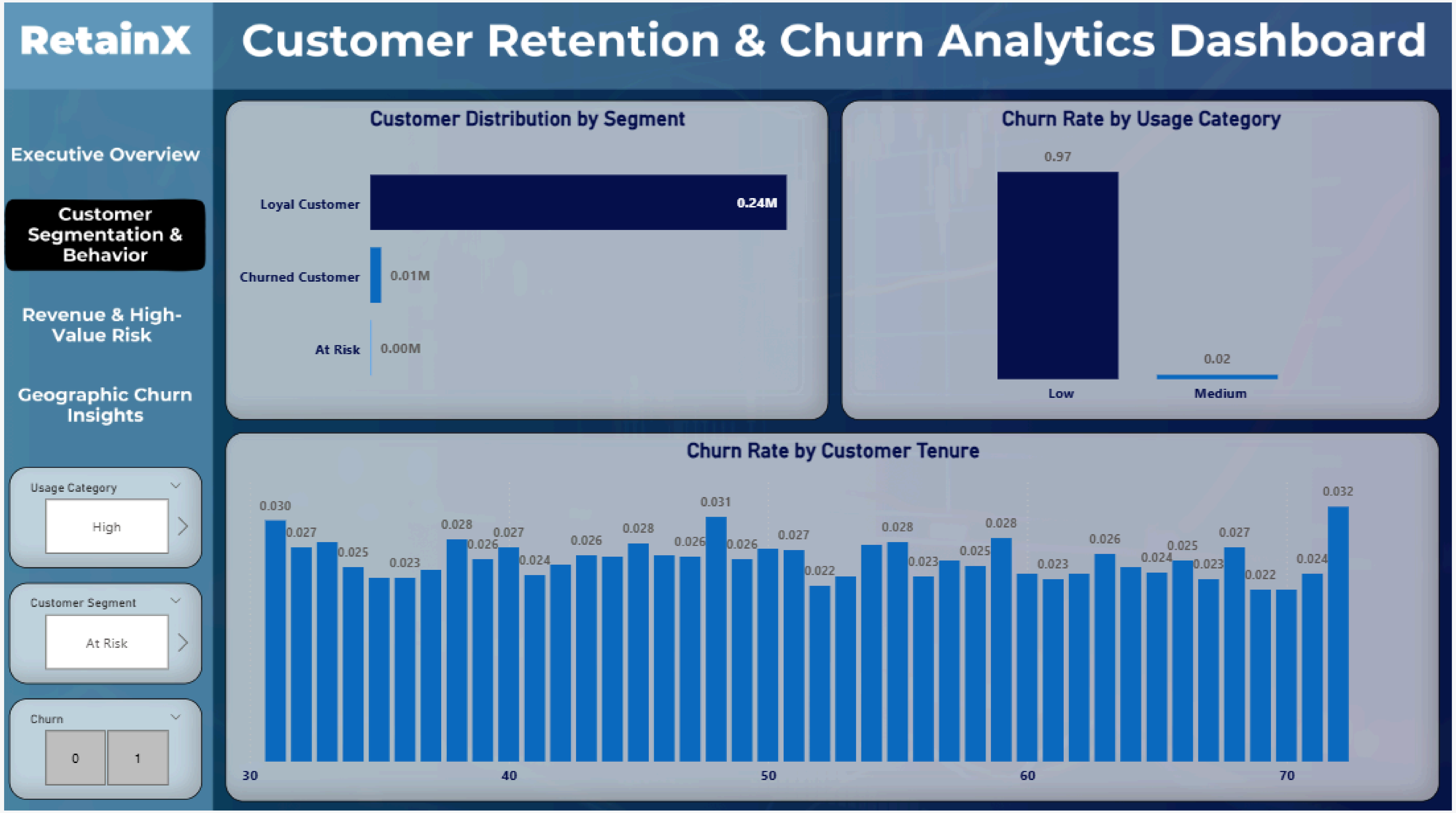
POWER BI DASHBOARD



POWER BI DASHBOARD



POWER BI DASHBOARD



KEY INSIGHTS SUMMARY

- What We Learned

1. Low usage is the strongest churn indicator
2. Longer tenure → higher retention
3. High-income churn causes disproportionate revenue loss
4. Geographic churn patterns are non-uniform
5. Early identification can prevent significant losses

BUSINESS RECOMMENDATIONS

ACTIONABLE NEXT STEPS

- Introduce usage-based retention offers
- Proactively target high-income, low-usage customers
- Launch region-specific churn campaigns
- Build early-warning churn flags using usage trends

THANK YOU